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Prototype of an Equity-Linked Structured Products Engine for Retail Investors

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Master Thesis

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Abstract

Retail participation in EU capital markets is comparatively low, with a large share of household wealth held in deposits; a key stated friction is that retail investors struggle to access information that is relevant, comparable and easily understandable. There is a lack of trust in investment advice and investors are exposed to inappropriately marketing information through conventional and new marketing channels. The European Commission proposed a retail investment strategy aiming to empower retail investors, foster trust, and improving comparability.

This thesis aims to contribute in this matter through the development of an interactive prototype with educational purpose that increases transparency and comparability between correlated multi-basket assets with exposure in different regions under a Geometric Brownian Motion framework, and equity-linked notes with the possibility of capital protection. The prototype also illustrates cross-border considerations through alternative foreign-exposure treatments (Compo versus Quanto). From the scenario based testing the prototype presents an histogram-based comparison that highlights the difference in return distribution between a buy and hold strategy and structured product strategies, showing how structured products reshape the payoff profile resulting in different return distributions, offering alternatives that could be attractive for different retail investors risk profiles. To reduce cognitive loads, the interface presents side by side strategy views, lightweight payoff plots that vary one underlying at a time and histogram comparison.

The thesis concludes that distribution-focused, comparison-first digital tools can operationalize policy goals around transparency and investor empowerment, while noting limitations from stylized assumptions (constant parameters, no issuer credit risk, fees, liquidity frictions, or path-dependent features) and outlining extensions toward more market-realistic implementations.

Executive Summary

Retail participation in European capital markets remains structurally low relative to other advanced economies, while a large fraction of household wealth stays concentrated in bank deposits. EU policy initiatives such as the Capital Markets Union and the Retail Investment Strategy explicitly link broader household participation to long-term objectives including retirement preparedness, diversification of funding sources for firms, and deeper capital markets. However, policy and academic work consistently highlight that increasing participation is not only a supply problem, i.e. more products, but also an information and trust problem. Retail investors struggle to access independent, comparable, and understandable information across products and channels, and this friction can be amplified by marketing incentives, heterogeneity in disclosures, and the cognitive burden of interpreting complex payoffs. A central recommendation in the European Commission's study on retail investor tools is the development of a standardized "comparison engine" and a centralized information hub that improves neutrality, comparability, and usability of product information.

This thesis contributes to that agenda through the development of a focused prototype: an interactive, distribution-based tool that compares a buy-and-hold investment to equity-linked structured products written on the same underlying basket. The core contribution is not a new pricing model, but an operational design that transforms the comparison problem into a reproducible, transparent workflow. Comparability is enforced by construction: the benchmark and structured strategies share the same basket, maturity, and simulated risk-factor scenarios, so differences in outcomes are attributable to payoff engineering rather than to inconsistent assumptions. The tool avoids ranking or recommending strategies and instead highlights trade-offs through payoff curves and terminal return distributions. This "neutral display" choice aligns with policy concerns about avoiding biased presentation and with behavioural evidence that investors often interpret risk through loss probability and asymmetry rather than variance alone.

The modelling framework combines tractability with enough structure to reflect the mechanisms that matter for cross-asset and cross-currency products. Each equity is modelled under a geometric Brownian motion with constant parameters. To reflect the reality that portfolios and structured underlyings often span regions and currencies, the tool simulates a unified set of correlated risk factors, including both equity shocks and foreign-

exchange shocks. Correlations are entered explicitly through a single factor-correlation matrix visible in the interface, and correlated shocks are generated via Cholesky factorization, with a numerical repair step if the user-defined matrix is not positive semi-definite. This choice supports auditability: the user can see, modify, and stress-test the dependence assumptions that drive joint outcomes.

A key design decision is the separation between simulation under the physical measure and valuation under the risk-neutral measure. Buy-and-hold outcomes are generated under $\mathbb{P}$ using expected returns as drifts for equity, while FX rates are simulated using interest-rate parity style drifts. Structured products, in contrast, are valued under $\mathbb{Q}$ using risk-free discounting and Black-style formulas. This dual-measure design improves interpretability: it reflects the practical distinction between "what outcomes could occur" (scenario distributions) and "how the option budget is priced" (no-arbitrage valuation). To support replicability for documentation and research use, the prototype fixes Monte Carlo settings in the thesis use case and initializes the random generator with a fixed seed, so that the same inputs lead to the same distributions and figures.

The structured products implemented are intentionally restricted to the most transparent class of equity-linked instruments: Type 1 products whose payoff remains directly linked to the underlying performance and can be decomposed into a funding leg and plain-vanilla option legs. In particular, the tool implements (i) a capital-protected participation note (ZCB + basket call), (ii) a capped participation note (ZCB + basket call spread), and (iii) a protected risk-reversal note (ZCB + long basket call + short basket put). Each payoff is defined directly as a function of the terminal basket level B_T, ensuring that the benchmark and structured strategies are expressed on the same object. The option leg is priced as an option *on the basket* rather than as a basket of separate options. Since a basket is a sum of lognormal variables and has no closed-form lognormal distribution in general, the tool uses a standard moment-matching approximation: the first two risk-neutral moments of B_T are computed and mapped into an effective lognormal proxy with an implied basket forward and volatility. Options are then priced in Black forward form, enabling fast interactive evaluation while preserving a consistent link between modelling assumptions and outputs.

The interface design follows a "comparison-first" principle intended to reduce cognitive load. All inputs are collected in a single sidebar, while the main view keeps the benchmark and the structured strategies visible at the same time. Rather than requiring users to navigate multiple tabs, the tool emphasizes immediate visual feedback: a path fan chart and return histogram for the benchmark, and side-by-side panels for structured strategies, each showing a payoff curve and a histogram comparison against buy-and-hold. This format makes it easy to isolate the impact of individual parameters while holding others constant. For example, reducing the capital protection level increases the option budget and typically increases participation, which is visible as a more "equity-like" payoff curve

and a wider terminal return distribution. Adjusting strike moneyness shifts payoff kinks and changes the distribution of outcomes in ways that can be observed immediately. This distribution-first presentation is aligned with behavioural findings that perceived risk is strongly related to the probability of loss and to downside asymmetry, and with the idea that payoff design can shape investor willingness to participate by limiting psychologically salient losses.

Overall, the thesis concludes that an interactive, distribution-focused prototype can operationalize key policy goals around transparency, comparability, and investor empowerment in a concrete and testable way. By binding every output to explicit assumptions (parameters, correlations, currency convention) and by keeping the benchmark and alternatives on the same footing, the tool illustrates how structured products reshape risk-return profiles relative to buy-and-hold. At the same time, the thesis emphasizes that results must be interpreted within the scope and simplifying assumptions of the prototype. Parameters are constant, issuer credit risk and funding spreads are not modelled, fees and transaction costs are not incorporated, and the product universe is limited to Type 1 structures without path dependence or early termination features. These abstractions are appropriate for an educational and comparability-focused prototype, but they imply that the tool is not a full product recommendation system and should not be interpreted as such.

Several extensions follow naturally from this work. A first direction is to increase market realism by incorporating fees, issuer credit risk proxies, and liquidity/early-exit considerations, which are central to real investor outcomes and to the interpretation of "capital protection" in practice. A second direction is to broaden the product universe toward more complex structures (e.g. autocallables or barrier features) while maintaining a transparency-first interface, potentially by introducing a gradual complexity indicator and modular explanations linked to payoff mechanics. More broadly, the prototype provides a scalable foundation for future development toward the type of centralized comparison infrastructure envisioned in EU policy: a tool where disclosures are not merely presented, but transformed into comparable, interpretable, and decision-relevant distributions.

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Notation

Table 1: Notation for stochastic modelling, basket construction, and simulation (Chapter 2).

Symbol	**Meaning**
t, T	Current time and maturity (in years)
Δt	Simulation time step
n	Number of assets in the basket
$i \in \{1, \ldots, n\}$	Asset index
m	Monte Carlo path index
$S_t^{(i)}$	Price of asset i at time t (in the asset's local currency)
$S_0^{(i)}$	Initial price of asset i
$X_t^{(i)}$	FX rate for asset i: domestic currency per 1 unit of asset-i currency at time t
$X_0^{(i)}$, $X_T^{(i)}$	FX rate at inception / at maturity (scenario-dependent if stochastic)
$\bar{X}$	Fixed FX conversion rate used in quanto contracts
$\bar{X}^{(i)}$	Fixed FX conversion rate for asset i (quanto conversion)
P_t	Domestic-currency underlying for a single foreign equity under a given convention
$P_t^{(i)}$	Domestic-currency price of asset i (domestic/compo/quanto case, depending on context)
$P_T^{(i)}$	Domestic-currency price of asset i at maturity, used in basket definition
V_0	Initial investment amount (domestic currency)
V_t^{BH}	Buy-and-hold basket value at time t (domestic currency)
$V_T^{BH,m}$	Terminal buy-and-hold basket value at maturity in path m (domestic currency)
w_i	Portfolio weight allocated to asset i, $w_i \geq 0$ and $\sum_{i=1}^n w_i = 1$
N_i	Number of shares/units of asset i held in the buy-and-hold basket
B_T	Basket value at maturity, $B_T = \sum_{i=1}^n N_i P_T^{(i)}$
B_0	Basket value at inception (domestic currency)
$\ln(\cdot)$	Natural logarithm
$R_T^{BH,(m)}$	Buy-and-hold simple return over $[0, T]$ in path m
$R_{T,\log}^{BH,(m)}$	Buy-and-hold log return over $[0, T]$ in path m
μ_i	Expected (physical-measure) return parameter for asset i
r	Continuously compounded risk-free rate (generic)
r_d	Domestic continuously compounded risk-free rate
r_f	Foreign continuously compounded risk-free rate (generic)
$r_{f,i}$	Foreign risk-free rate corresponding to currency of asset i
q, q_i	Continuous dividend yield of the underlying (generic / for asset i)
σ, σ_i	Equity volatility (generic / for asset i in its local currency)
σ_S	Equity volatility (when emphasizing equity vs FX in a two-factor setting)
σ_X, σ_{X_i}	FX volatility (generic / for currency i)
σ_P	Effective domestic volatility of $P_t = S_t X_t$ in the Compo case
$\sigma_{P,i}$	Domestic log-volatility of $P^{(i)}$ used in basket moment matching
ρ_{ij}	Correlation between domestic log-returns (or Brownian shocks) of assets i and j
ρ_{SX}	Correlation between equity and FX shocks in the Compo/Quanto setting
$\mathbf{C}$	Correlation matrix of risk-factor shocks (equities and FX)
$\mathbf{L}$	Cholesky factor of $\mathbf{C}$ such that $\mathbf{C} = \mathbf{L}\mathbf{L}^\top$
$\mathbf{W}_t$	Vector of Brownian motions driving the simulated risk factors

Table 2: Notation for stochastic modelling, basket construction, and simulation (Chapter 2).

Symbol	**Meaning**
W_t	Standard Wiener process (Brownian motion)
$\boldsymbol{\varepsilon}_k$	Vector of independent standard normal draws at time step k
$\mathbf{Z}_k$	Correlated normal shock vector at time step k, $\mathbf{Z}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{C})$
$Z_{k,Y}$	Component of $\mathbf{Z}_k$ used for process Y at time step k
$\mathbb{P}$, $\mathbb{Q}$	Physical measure / risk-neutral measure
K	Option strike (in domestic currency unless specified otherwise)
K_1, K_2	Lower/upper strikes in a call spread ($K_2 > K_1$)
K_c, K_p	Call/put strikes in a risk-reversal payoff
C_0, P_0	Time-0 value of a European call / put
$N(\cdot)$	Standard normal cumulative distribution function
d_1, d_2	Black–Scholes / Black auxiliary terms
F_0	Forward price at $t = 0$ (generic)
$F_{P,0}^{\text{compo}}$	Domestic forward of $P_T = S_T X_T$ under Compo conversion
$F_{P,0}^{\text{quanto}}$	Domestic forward of $P_T = \bar{X} S_T$ under Quanto conversion
$F_{P,0}^{(i)}$	Domestic forward of $P_T^{(i)}$ used in basket moment matching
df	Discount factor (typically e^{-rT} or $e^{-r_d T}$)
F	Face value / protected amount paid by the ZCB at maturity
$PV_{ZCB}(0)$	Present value of a zero-coupon bond at time 0
α	Participation factor / number of option units per reference exposure
M_1, M_2	First and second moments of B_T under $\mathbb{Q}$, $M_1 = \mathbb{E}^{\mathbb{Q}}[B_T]$, $M_2 = \mathbb{E}^{\mathbb{Q}}[B_T^2]$
F_B	Basket forward proxy, $F_B = M_1$
σ_B	Basket lognormal proxy volatility from moment matching
$\mathcal{N}$	Note notional (used in the reverse convertible payoff)

Chapter 1

Introduction

1.1 Aims of the study

This thesis contributes a focused prototype that operationalizes several core principles for the study´s hub proposed solution (European Commission, 2020). While narrower scope, the prototype targets a specific class of products—equity-linked structured payoffs. It addresses the retail comparison problem through specific design choices that align with official recommendations regarding comparability, neutrality, and usability.

Comparability is enforced by construction. The tool places a buy-and-hold investment in the same underlyings as the structured product at the centre of the analysis, then compares it directly to multiple structured strategies under the same horizon and simulated market dynamics. This responds to the study's finding that the main challenge is effective comparison of key product features (risk, performance, cost) between alternatives, aligns with the "comparison engine" that the hub is meant to enable (European Commission, 2020).

The tool improves interpretability by translating product design into mechanisms and distributions. It addresses the struggle of gathering independent, comparable information by decomposing each structured payoff into a transparent allocation between capital protection and an options budget. Moreover, rather than relying solely on static descriptions, the tool presents payoff curves and terminal distributions. This approach aligns with the hub's goal of improving the practical usability of disclosed information to help investors make well-informed decisions.

The interface choices reflect the study's emphasis on neutrality and managing cognitive load. To ensure this, the tool avoids ranking the various strategies; instead, it presents visualized trade-offs and leaves the final evaluation to the user. Additionally, the tool follows a "display priority logic" that shows one default payoff while allowing users to vary the options. This reduces complexity without hiding the underlying structure, echoing the study's warning that presenting too many data elements can overwhelm

users.

1.2 Policy landscape: why retail participation matters in the EU

In the European Union, retail participation in capital markets remains structurally low compared to other advanced economies. The EU council notes that in 2021, only about 17% of EU household assets were held in financial securities versus 43% held in the US, while a large share of EU households wealth remains in bank deposits despite the historical gains associated with long term equity investing (Council of the European Union, 2023).

The European Union's Capital Market Union (CMU) agenda aims to invigorate retail investor participation in financing the economy. By putting European savings to better use, the initiative ensures that EU citizens can meet long-term objectives, such as preparing for retirement or saving for unexpected expenses. Furthermore, the CMU seeks to improve EU capital markets by increasing the borrowers and savers match making. This process increases market depth and provides essential funding diversification (European Commission, 2020).

The Retail Investment Strategy explicitly frames its objectives as increasing consumers protection and trust to encourage investment in EU capital markets. To achieve this, it highlights several recurring frictions that currently prevent households from benefiting from these markets. Common obstacles include the difficulty of accessing relevant, comparable and understandable information and a general lack of confidence in investment advice. This mistrust is further exacerbated by the risk of unrealistic marketing information on social media and new marketing channels and the high fees associated with certain services, which ultimately diminish investment returns. In response, reforms aims to improve digital accessibility and comparability of product information. Specifically, updates through updates to PRIIPs Key Information Documents (KIDs) and cost/risk disclosure rules, allowing investors a better understanding and compare products (Council of the European Union, 2023).

Although the regulation expanded disclosure requirements, disclosure alone does not automatically translate into usable understanding. A study on online tools for retail investment decisions argues that retail investors face significant challenges when comparing product features across different distribution channels. While digital solutions have the potential to improve how information is presented, current online tools are often unsatisfactory. Many of these tools only cover a small portion of the product universe, lack transparency and independence, or are restricted behind paywalls. To address these issues, the study proposes the creation of a centralized investor products hub. This database would provide retail investors with free access to complete, accurate, and in-

dependent information on managed financial products across all EU markets (European Commission, 2020).

1.3 Structured products focus

The thesis adopts the view that increasing retail participation is not simply a matter of offering more products. Instead, it requires a better alignment between household concerns and market offerings, all while ensuring risks remain transparent.

A key behavioural challenge is that retail investors often do not evaluate risk the way models assume. Holzmeister et al. (2020) show through a large international survey experiment with finance professionals and laypeople that skewness, and especially loss probability, are strong drivers of perceived risk. In contrast, variance can be surprisingly weak in shaping how people perceive danger. In other words, many investors instinctively anchor on “How likely am I to lose money?” and “Do I have upside?” more than on standard deviation. This distinction is critical. Plain buy-and-hold equity exposure often appears psychologically unattractive when framed as an uncertain outcome with a meaningful downside probability. This is especially true for households whose savings are currently held in "safe" bank deposits (Holzmeister et al., 2020).

Relatedly, disappointment based preferences can rationalize limited stock market participation even when equity premia exist.The work of Ang et al. (2005) on disappointment aversion demonstrates that if investors are more sensitive to outcomes falling below a reference point (disappointment) than to upside, optimal choices can include include low equity allocations or even non-participation. This helps explain why historically given gains by stocks may be an insufficient argument for many households (Ang et al., 2005).

Against this backdrop, structured products are not merely "complex retail instruments". properly designed and explained, they can be interpreted as security design that reshapes the payoff distribution to match household loss concerns while preserving access to market upside. Calvet et al. (2023) provide empirical evidence consistent with this view.Their study found that the introduction and widespread adoption of capital-guarantee products in Sweden was associated with an increase in expected portfolio returns. The effects were particularly strong for households that were initially less willing to take risks. These findings suggest that specific payoff designs can foster household risk-taking by mitigating behavioral frictions, such as loss aversion and narrow framing (Calvet et al., 2023).

1.3.1 Structured Products Selection

Structured products are bespoke instruments that enable investors to pursue strategies tailored to their market views and preferences. They can be engineered on a wide range

of underlyings, including equities, interest rates, foreign exchange, commodities, credit, or real estate. The key idea is that a structured payoff is not "mysterious": it is typically constructed by combining a funding component (often a zero-coupon bond that anchors repayment at maturity) with one or more derivative components (options) that shape the upside and downside exposure. This design flexibility creates a large universe of possible payoff profiles, but it also increases the risk of misunderstanding when the product mechanisms are not explained in payoff-based terms (Chia, 2010).

To bring order to this wide design space,Vandenbroucke (2014) proposes a pragmatic classification of structured products by benchmarking them against European plain-vanilla call and put options, treated as the basic "building blocks" of structured payoffs. The classification is intended to help navigate heterogeneous products within a limited number of types and can serve as a driver for a gradual complexity indicator. Products are ordered by the extent to which their characteristics remain close to plain-vanilla option logic, using three option-like criteria that are applied strictly and sequentially: (i) whether embedded options have a fixed maturity, (ii) whether they reference a single underlying, and (iii) whether the payoff is directly linked to the underlying's performance. Applying these criteria in strict order yields four product types (Vandenbroucke, 2014).

- **Type 1:** Closest to plain-vanilla logic; includes call/put options by definition and "bond $\pm$ option" notes where the payoff is directly linked to underlying performance (possibly capped).
- **Type 2:** Fixed maturity and single underlying, but payoff includes additional mechanisms such as averaging (Asian), barriers, digital effects, lookbacks, or volatility-targeting overlays, which weaken the simple point-to-point link to S_T.
- **Type 3:** Basket structures where the payoff depends on individual constituent performances, making correlation structurally relevant for valuation and payoff intuition.
- **Type 4:** Products with early termination features (e.g., autocallables/callables/puttables), moving furthest away from the plain-vanilla building blocks.

For the sake of scope and time, this thesis focuses on Type 1 equity-linked structures (Vandenbroucke, 2014). This choice is not merely a simplification; it is aligned with the research objective of improving transparency and interpretability for retail investors through an interactive tool. Type 1 products are the most suitable starting point because their payoffs can be explained modularly using a small number of mechanisms (bond funding plus plain-vanilla option legs) and can be priced within a coherent and transparent framework. In contrast, Types 2-4 introduce path-dependence (e.g., averaging and barriers), multi-dimensional dependence (basket constituents and correlation effects), and

early termination features that substantially increase both modelling requirements and user-interface complexity, which would dilute the core contribution of this thesis (Vandenbroucke, 2014). By focusing on Type 1, the tool can prioritise clarity while still covering the essential retail trade-off between capital protection, upside participation, capped participation, and financed upside via downside exposure, thereby providing a meaningful and scalable foundation for later extensions (Vandenbroucke, 2014).

1.4 Link to Sustainable Development Goals (SDG´s)

This project can be linked to the 8th goal: Decent work and economic growth (United Nations Development Programme, 2015). A deeper and more inclusive capital market supports long-run growth by improving how savings are channelled into productive investment. In the EU context, the CMU objective of putting European savings to better use aligns with strengthening economic resilience and financing capacity, including for firms that rely on market-based funding rather than only bank credit (Council of the European Union, 2023). At the same time, 9th goal: Industry, innovation and infrastructure, is relevant because this thesis is explicitly about building digital financial infrastructure at the user level: transforming complex disclosures into an interactive tool that improves access to understandable investment information (United Nations Development Programme, 2015). Reducing opacity and enabling a clearer comparison between buy-and-hold and structured payoffs, the tool supports more informed participation and helps reduce the likelihood that households remain excluded from capital market benefits due to avoidable information barriers, an inclusion channel that complements the broader growth and innovation goals of the CMU agenda (Council of the European Union, 2023; United Nations Development Programme, 2015).

Chapter 2

Theoretical Background

This chapter summarizes the financial theory underpinning the tool developed in this thesis and presents the core equations that will be referenced in Chapter 4 when describing the implementation. Following Hull (2018) and Oesterreichische Nationalbank (2004), we first present the stochastic framework used for equity dynamics and the risk-neutral valuation setting. We then extend the single-asset model to a multi-asset basket with correlation and joint simulation. Next, we present the fixed-income and option-pricing building blocks used to construct equity-linked payoffs. Finally, because some underlyings may be denominated in foreign currencies, we describe how FX risk enters the payoff and how it can be treated through Compo (unhedged) or Quanto (hedged) conventions.

2.1 Stochastic modelling of equity prices

In order to generate equity scenarios for pricing and payoff analysis, the tool models each stock (or equity proxy) as a continuous-time stochastic process. Following the Black–Scholes–Merton framework in Hull (2018), prices are assumed to evolve according to a geometric Brownian motion (GBM), which is a particular case of an Itô process to which Itô's lemma can be applied (Hull, 2018).

Information flow and modelling perspective

A common modelling implication of the weak form of market efficiency is that the current price incorporates the information contained in past prices, so that past trajectories should not add systematic predictive power beyond the current state. This motivates adopting a Markov modelling framework, in which the conditional distribution of future values depends on the current level of the process rather than on the full past history (Hull, 2018).

At the same time, efficiency is naturally interpreted as limiting predictability of innovations in returns: after conditioning on current information, the remaining shock

component is assumed to have zero conditional mean (a martingale-difference type condition). This perspective supports modelling price changes as driven by sequences of unpredictable shocks that are independent across non-overlapping time intervals, leading to random-walk and diffusion-type specifications (Hull, 2018).

Gaussian random walk intuition

A convenient discrete-time starting point is a Gaussian random walk for a state variable $(X_k)_{k\geq 0}$ (often interpreted as a log-price), defined by:

$$X_{k+1} = X_k + \varepsilon_{k+1}, \qquad \varepsilon_k \overset{i.i.d.}{\sim} \mathcal{N}(0,1). \tag{2.1}$$

The one-step increment is therefore:

$$\Delta X \equiv X_{k+1} - X_k = \varepsilon_{k+1}. \tag{2.2}$$

Over m steps, the cumulative change is a sum of m independent shocks:

$$X_{k+m} - X_k = \sum_{j=1}^{m} \varepsilon_{k+j},$$

implying $\mathbb{E}(X_{k+m} - X_k) = 0$ and $\mathrm{Var}(X_{k+m} - X_k) = m$ (Hull, 2018).

2.1.1 Wiener process (Brownian motion)

Continuous-time limit and scaling

Moving to continuous time can be viewed as taking a diffusion limit of the discrete random walk. The key assumptions are (i) increments over non-overlapping time intervals are independent, and (ii) the variance of the increment scales linearly with the length of the interval. This leads to a standard Wiener process $(W_t)_{t\geq 0}$ such that, for $0 \leq s < t$:

$$W_t - W_s \sim \mathcal{N}(0,\, t-s), \tag{2.3}$$

and increments over disjoint intervals are independent (Hull, 2018). On a discrete time grid $t_k = k\Delta t$, this is commonly represented as:

$$W_{t_{k+1}} - W_{t_k} = \sqrt{\Delta t}\,\varepsilon_{k+1}, \qquad \varepsilon_k \overset{i.i.d.}{\sim} \mathcal{N}(0,1). \tag{2.4}$$

where $\mathbb{E}(W_{t_{k+1}} - W_{t_k}) = 0$, $\mathrm{Var}(W_{t_{k+1}} - W_{t_k}) = \Delta t$, and $\mathrm{Std}(W_{t_{k+1}} - W_{t_k}) = \sqrt{\Delta t}$ (Hull, 2018).

Over $N = \frac{T}{\Delta t}$ steps, the increment can be written as a sum of independent one-step

increments:

$$W_T - W_0 = \sum_{j=1}^{N} \sqrt{\Delta t}\, \varepsilon_j. \tag{2.5}$$

Hence $\mathbb{E}(W_T - W_0) = 0$, $\mathrm{Var}(W_T - W_0) = T$, and $\mathrm{Std}(W_T - W_0) = \sqrt{T}$. Brownian motion is also Markov; in this setting the Markov property is consistent with the independent-increments construction (Hull, 2018).

Generalized Wiener process (arithmetic Brownian motion)

A flexible extension is obtained by adding a deterministic drift and scaling the diffusion term:

$$dX_t = a\, dt + b\, dW_t, \tag{2.6}$$

where $a \in \mathbb{R}$ and $b > 0$ are constants. Over an infinitesimal interval dt, this implies $\mathbb{E}(dX_t) = a\, dt$ and $\mathrm{Var}(dX_t) = b^2\, dt$, and over a horizon $[0, T]$:

$$\mathbb{E}(X_T - X_0) = aT, \qquad \mathrm{Var}(X_T - X_0) = b^2 T.$$

Equation (2.6) is also known as arithmetic Brownian motion (Hull, 2018).

2.1.2 Geometric Brownian motion (GBM)

As described in Hull (2018), let $(S_t)_{t\in[0,T]}$ denote the equity price process. In general, an Itô process for a state variable X_t can be written as:

$$dX_t = a(t, X_t)\, dt + b(t, X_t)\, dW_t, \tag{2.7}$$

where $a(t, X_t)$ is the drift term, $b(t, X_t)$ controls the magnitude of random fluctuations, and W_t is a standard Wiener process. To model stock prices, it is natural to assume that both the expected change and the uncertainty scale proportionally with the current price level. This leads to the GBM specification:

$$dS_t = \mu S_t\, dt + \sigma S_t\, dW_t, \tag{2.8}$$

where we adopt time-homogeneous GBM with constant drift (expected return: μ) and diffusion (volatility: σ) as a parsimonious baseline that yields normally distributed log returns and a tractable lognormal price distribution (Hull, 2018).

2.1.3 Itô's lemma and log-price dynamics

Itô's lemma describes how a sufficiently smooth function of an Itô process evolves. We introduce it here because it allows us to transform the GBM dynamics of S_t into tractable

dynamics for functions of S_t (in particular $\ln S_t$), which yields log-return normality and forms the analytical basis for both Monte Carlo simulation and Black–Scholes–Merton valuation (Hull, 2018). If S_t follows (2.8) and $G(t, S)$ is twice differentiable in S and once in t, then $G(t, S_t)$ satisfies (Hull, 2018):

$$dG = \left(\frac{\partial G}{\partial t} + \mu S_t \frac{\partial G}{\partial S} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 G}{\partial S^2} \right) dt + \sigma S_t \frac{\partial G}{\partial S}\, dW_t. \tag{2.9}$$

To obtain log-price dynamics, we choose:

$$G(S) = \ln S. \tag{2.10}$$

Then $\frac{\partial G}{\partial S} = \frac{1}{S}$ and $\frac{\partial^2 G}{\partial S^2} = -\frac{1}{S^2}$, and applying (2.9) yields:

$$d \ln S_t = \left(\mu - \frac{1}{2}\sigma^2 \right) dt + \sigma\, dW_t. \tag{2.11}$$

2.1.4 Log returns and lognormality

The continuously compounded (log) return over $[0, T]$ is defined by

$$R_T^{\log} = \ln \left(\frac{S_T}{S_0} \right). \tag{2.12}$$

Integrating (2.11) over $[0, T]$ implies that $\ln S_T$ is normally distributed:

$$\ln S_T \sim \mathcal{N}\left(\ln S_0 + \left(\mu - \frac{1}{2}\sigma^2 \right) T,\ \sigma^2 T \right). \tag{2.13}$$

Equivalently, $\ln S_T - \ln S_0$ is normally distributed (Hull, 2018).

2.1.5 Risk-neutral dynamics and dividends

For no-arbitrage pricing of derivatives, valuation is performed under the risk-neutral measure $\mathbb{Q}$, under which the expected growth rate of a dividend-paying stock equals the risk-free rate net of the continuous dividend yield q, i.e. $dS_t/S_t = (r - q)\, dt + \sigma\, dW_t^{\mathbb{Q}}$ (Hull, 2018). This is the standard Black–Scholes–Merton argument explaining why option prices depend on r and q rather than on the real-world expected return μ.

By contrast, buy-and-hold performance simulations are naturally described under the physical measure $\mathbb{P}$, where the expected return is governed by a drift parameter μ (depending on convention, μ may refer to total expected return, in which case the ex-dividend price drift is $(\mu - q)$).

2.1.6 Discretization and Monte Carlo simulation

To simulate paths, the tool uses the exact log-discretization associated with (2.13) under constant parameters. For a grid $t_k = k\Delta t$ and i.i.d. $\varepsilon_k \sim \mathcal{N}(0,1)$:

Physical-measure (performance) simulation

$$\ln S_{t_{k+1}} = \ln S_{t_k} + \left((\mu - q) - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t}\,\varepsilon_k, \tag{2.14}$$

Risk-neutral (pricing) simulation

$$\ln S_{t_{k+1}} = \ln S_{t_k} + \left((r - q) - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t}\,\varepsilon_k, \tag{2.15}$$

and $S_{t_{k+1}} = \exp(\ln S_{t_{k+1}})$. This discretization is the basis for the Monte Carlo distributions shown later in the tool outputs (Hull, 2018).

2.2 Multi-asset baskets

This section defines the buy-and-hold basket used as the benchmark in the tool and introduces the currency conventions required to express basket values from the domestic investor perspective. Because the basket may include foreign denominated assets, the domestic value is exposed to FX risk.

2.2.1 Basket definition and weights

Consider n risky assets with price processes $S_t^{(i)}$, each quoted in its *own* (local) currency. At inception $t = 0$, the investor selects portfolio weights $w_i \geq 0$ such that $\sum_{i=1}^{n} w_i = 1$. The tool treats these weights as buy-and-hold allocations that remain fixed in terms of initial capital invested in each asset (Hull, 2018).

2.2.2 Domestic valuation and FX conventions

Let $X_t^{(i)}$ denote the exchange rate quoted as *domestic currency per one unit of the currency of asset i*. By convention, domestic assets have $X_t^{(i)} \equiv 1$. The domestic-currency unit price of asset i at inception is therefore $S_0^{(i)} X_0^{(i)}$. Given initial capital V_0, the number of shares purchased in asset i is:

$$N_i = \frac{w_i V_0}{S_0^{(i)} X_0^{(i)}}. \tag{2.16}$$

For any time t, the domestic-currency buy-and-hold basket value is:

$$V_t^{BH} = \sum_{i=1}^{n} N_i \, S_t^{(i)} X_t^{(i)}. \tag{2.17}$$

Consequently, for each simulation path m, the terminal basket value at maturity T is:

$$V_T^{BH,m} = \sum_{i=1}^{n} N_i \, S_T^{(i,m)} X_T^{(i,m)}. \tag{2.18}$$

2.2.3 Basket returns

For each path m, the buy-and-hold simple return and log return over $[0, T]$ are defined as (Hull, 2018):

$$R_T^{BH,(m)} = \frac{V_T^{BH,(m)}}{V_0} - 1, \qquad R_{T,\log}^{BH,(m)} = \ln\left(\frac{V_T^{BH,(m)}}{V_0}\right). \tag{2.19}$$

Finally, it is important to note that even if each $S_T^{(i)}$ is lognormal, the basket value is a *weighted sum of lognormals*, which is not lognormal in general (Hull, 2018). Consequently, the distribution of basket log-returns is not exactly normal.

2.2.4 Correlation structure and joint simulation

In order to model correlated movements across assets and currencies, the tool simulates a joint vector of Brownian shocks (Hull, 2018). Let $\mathbf{W}_t$ collect the Brownian drivers of all simulated risk factors, including:

- equity shocks for each asset,
- FX shocks for each foreign currency,
- cross-dependence between equities and FX rates.

Let $d\mathbf{W}_t$ be a d-dimensional Brownian increment vector with correlation matrix $\mathbf{C} \in \mathbb{R}^{d \times d}$, where $\mathbf{C}$ contains the relevant pairwise correlations between the above risk-factor shocks. By construction, $C_{ii} = 1$ and $C_{ij} = \rho_{ij}$ (Hull, 2018).

Cholesky factorization and correlated normals

To generate correlated standard normal shocks on a discrete time grid, the tool uses a Cholesky factorization:

$$\mathbf{C} = \mathbf{L}\mathbf{L}^\top, \tag{2.20}$$

where $\mathbf{L}$ is lower triangular. At each time step $t_k \to t_{k+1}$, draw a vector $\boldsymbol{\varepsilon}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ and set:

$$\mathbf{Z}_k = \mathbf{L}\boldsymbol{\varepsilon}_k. \tag{2.21}$$

Then $\mathbf{Z}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{C})$ and provides the correlated shocks used in the discretization of all simulated processes (Hull, 2018).

Practical considerations

The correlation matrix $\mathbf{C}$ must be positive semidefinite for (2.20) to exist; in practice, standard numerical Cholesky routines require a positive definite matrix, so empirical estimates may require minor regularization for stability.

2.2.5 Dynamics under correlated processes

To ensure accuracy in returns computations, the tool generates a common set of correlated equity and FX scenarios that shows the effect of exchange rate risk in the portfolio performance when the basket includes foreign currencies.

Equity processes In the scenario engine, each equity follows a GBM driven by its own Brownian shock. Under the physical measure $\mathbb{P}$ (performance scenarios), the equity dynamics are written as:

$$\frac{dS_t^{(i)}}{S_t^{(i)}} = (\mu_i - q_i)\,dt + \sigma_i\,dW_t^{(S_i)}, \qquad i = 1, \ldots, n, \tag{2.22}$$

where μ_i is the expected return, q_i is the continuous dividend yield, and σ_i the equity volatility (Hull, 2018). When risk-neutral inputs are needed (e.g. for forward prices in option valuation), μ_i is replaced by the appropriate risk-neutral drift $(r - q_i)$.

FX processes For each foreign currency i, the FX rate is modelled as:

$$\frac{dX_t^{(i)}}{X_t^{(i)}} = (r_d - r_{f,i})\,dt + \sigma_{X_i}\,dW_t^{(X_i)}, \qquad i \in \mathcal{F}, \tag{2.23}$$

where r_d is the domestic risk-free rate, $r_{f,i}$ the foreign risk-free rate for currency i, and σ_{X_i} the FX volatility (Chan et al., 2024). The drift in (2.23) corresponds to interest-rate parity under the domestic risk-neutral measure and is used in the tool as a parsimonious benchmark for FX scenario generation.

Exact discretization for simulation

On a grid $t_k = k\Delta t$, the tool uses the exact log-discretization of GBM. For any process Y_t satisfying $dY_t/Y_t = \mu_Y dt + \sigma_Y dW_t$, the update is:

$$\ln Y_{t_{k+1}} = \ln Y_{t_k} + \left(\mu_Y - \frac{1}{2}\sigma_Y^2\right)\Delta t + \sigma_Y\sqrt{\Delta t}\, Z_{k,Y}, \tag{2.24}$$

where $Z_{k,Y}$ is the correlated standard normal shock associated with Y at step k.

Correlated shocks are generated using the Cholesky construction described in Section 2.2.4. If $\mathbf{C}$ is the correlation matrix of all risk-factor shocks (equities and FX), $\mathbf{C} = \mathbf{L}\mathbf{L}^\top$ and $\mathbf{Z}_k = \mathbf{L}\boldsymbol{\varepsilon}_k$ with $\boldsymbol{\varepsilon}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ (Hull, 2018).

Equity processes

$$\ln S_{t_{k+1}}^{(i)} = \ln S_{t_k}^{(i)} + \left((\mu_i - q_i) - \frac{1}{2}\sigma_i^2\right)\Delta t + \sigma_i\sqrt{\Delta t}\, Z_{k,S_i}. \tag{2.25}$$

FX processes

$$\ln X_{t_{k+1}}^{(i)} = \ln X_{t_k}^{(i)} + \left((r_d - r_{f,i}) - \frac{1}{2}\sigma_{X_i}^2\right)\Delta t + \sigma_{X_i}\sqrt{\Delta t}\, Z_{k,X_i}. \tag{2.26}$$

2.3 Building blocks of equity-linked structured products

Equity-linked structured products can be engineered through different replication mechanisms and legal wrappers (Chia, 2010). At the payoff-engineering level, many designs can be decomposed into (i) a fixed-income component, used to deliver principal repayment, partial protection, or simply funding/discounting, and (ii) an equity-linked component, providing contingent exposure to the underlying. Depending on the design, the option leg may enhance upside participation, monetise downside risk, generate yield, or implement asymmetric exposures; therefore, the fixed-income part is not necessarily fully capital-protecting and the option part does not only "guide the upside" (Vandenbroucke, 2014). In practice, the capital-preserving component may be implemented in multiple ways: via a static allocation to fixed-income instruments (e.g. zero-coupon bonds), via secured or collateralized arrangements, or synthetically through derivatives (e.g. swap-based exposure). In addition, market practice distinguishes between products marketed as *capital guaranteed* and those marketed as *capital protected*, where the investor's effective protection depends not only on interest rates but also on design-specific risks such as issuer default risk, counterparty risk, liquidity risk, fees, and funding spreads. For instance, in the case of bank-issued structured notes, any principal repayment ultimately remains

exposed to issuer creditworthiness (Chia, 2010).

In this thesis, the objective is to compare payoff profiles and risk–return trade-offs across equity-linked strategies under a unified and transparent valuation framework. Since the replication route (bond holdings, secured structures, or swap-based implementation) is typically not observable to the end investor at the level of payoff diagrams, and because these alternatives mainly differ by the level and nature of the additional risks (default, liquidity, and funding spreads), we adopt an idealized decomposition into a default-free fixed-income leg and a derivative leg. Concretely, the capital-preserving component is modelled as a default-free zero-coupon bond discounted at the risk-free rate, which provides a clean benchmark for the protected amount at maturity. This abstraction isolates payoff engineering from issuer- and implementation-specific frictions; consequently, credit and liquidity premia are not priced explicitly and are discussed as limitations.

Given the focus on Vandenbroucke (2014) *Type 1* products whose payoffs remain close to plain vanilla call/put logic, the equity-linked component is modelled using European options priced in the Black–Scholes–Merton framework (Vandenbroucke, 2014). This choice ensures closed-form valuation, clear parameter interpretation, and computational efficiency, which is consistent with the tool's objective of allowing users to compare alternative payoff structures on a common basis (European Commission, 2020).

2.3.1 Fixed income building block: zero-coupon bonds

A zero-coupon bond (ZCB) paying face value F at maturity T has present value under continuous compounding:

$$PV_{ZCB}(0) = Fe^{-rT}. \tag{2.27}$$

In the tool, capital protection is implemented by allocating part of the initial budget to a ZCB (Hull, 2018).

2.3.2 Basket-linked option legs

A European call option gives the holder the right (but not the obligation) to buy the underlying at maturity T for strike K, with payoff $(S_T - K)^+$. A European put option gives the right to sell the underlying at T for strike K, with payoff $(K - S_T)^+$ (Hull, 2018). Short positions carry the corresponding obligations; combinations of calls and puts allow tailoring payoff profiles to specific market views (Hull, 2018).

Black–Scholes–Merton closed-form option pricing

The tool prices European options using the Black–Scholes–Merton (BSM) closed-form formula. Under GBM, stock prices have lognormal terminal distributions (Section 2.1.4),

which makes payoffs of the form $(S_T - K)^+$ and $(K - S_T)^+$ analytically tractable (Hull, 2018).

For valuation, the relevant dynamics are those under the risk-neutral measure $\mathbb{Q}$: in an arbitrage-free setting, derivative prices are obtained by discounting expected payoffs at the risk-free rate. Under $\mathbb{Q}$, the drift of the equity price is not the real-world expected return μ but the risk-free rate adjusted for dividends, $(r-q)$. Hence option values depend on (r, q, σ) and the contract terms (K, T), rather than on investors' return expectations (Hull, 2018).

The BSM formulas for t_0 values of a European call and put are as follows (Hull, 2018):

$$C_0 = S_0 e^{-qT} N(d_1) - K e^{-rT} N(d_2), \tag{2.28}$$

$$P_0 = K e^{-rT} N(-d_2) - S_0 e^{-qT} N(-d_1), \tag{2.29}$$

with (Hull, 2018):

$$d_1 = \frac{\ln(S_0/K) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}. \tag{2.30}$$

Forward representation: Black's model

In implementation it is convenient to work with the forward price:

$$F_0 = S_0 e^{(r-q)T}. \tag{2.31}$$

European options written on spot and European options written on a forward (or futures) are equivalent when the option matures at the same time as the forward (or futures) (Hull, 2018).

The same option values can be written in Black forward form (Hull, 2018):

$$C_0 = e^{-rT} \left[F_0 \, N(d_1) - K \, N(d_2) \right], \tag{2.32}$$

$$P_0 = e^{-rT} \left[K \, N(-d_2) - F_0 \, N(-d_1) \right], \tag{2.33}$$

where (Hull, 2018):

$$d_1 = \frac{\ln(F_0/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}. \tag{2.34}$$

2.3.3 Foreign underlyings: Compo and Quanto conventions

The formulas above price options written on an underlying already expressed in domestic currency. If an equity is quoted in a foreign currency, the domestic payoff depends on the FX conversion rule. Let S_t denote the foreign equity price in its own currency and

let X_t denote the FX rate quoted as domestic currency per one unit of foreign currency (Oesterreichische Nationalbank, 2004).

To unify notation, we define the *domestic-currency underlying* process $(P_t)_{t\in[0,T]}$ as the foreign equity price expressed in domestic currency under the chosen conversion convention:

$$P_t \equiv \begin{cases} S_t X_t, & \text{Compo (unhedged)}, \\ \bar{X}\, S_t, & \text{Quanto (hedged conversion)}, \end{cases} \tag{2.35}$$

where $\bar{X}$ is a fixed conversion rate agreed at inception (Oesterreichische Nationalbank, 2004). European options are then priced as options on P_T in Black forward form, using the domestic forward $F_{P,0}$ and the appropriate volatility input.

Compo adjustment (option on $P_T = S_T X_T$)

Under the standard lognormal approximation for (S_t, X_t), the domestic underlying $P_t = S_t X_t$ is lognormal and has an effective volatility (Oesterreichische Nationalbank, 2004):

$$\sigma_P = \sqrt{\sigma_S^2 + \sigma_X^2 + 2\rho_{SX}\sigma_S\sigma_X}, \tag{2.36}$$

where σ_S is equity volatility in foreign currency, σ_X is FX volatility, and ρ_{SX} is the equity–FX correlation.

In forward form, the domestic forward of P_T under domestic discounting is:

$$F_{P,0}^{\text{compo}} = S_0 X_0\, e^{(r_d - q)T}, \tag{2.37}$$

and the option is priced via (2.32)–(2.33) using $(F_{P,0}^{\text{compo}}, \sigma_P)$ and domestic discounting $e^{-r_d T}$ (Oesterreichische Nationalbank, 2004).

Quanto adjustment (option on $P_T = \bar{X} S_T$)

Under quanto conversion, FX randomness is removed from the payoff, but valuation incorporates the equity–FX dependence through an adjustment to the domestic forward (change-of-numeraire argument) (Oesterreichische Nationalbank, 2004). A convenient representation for the domestic forward of $P_T = \bar{X} S_T$ is:

$$F_{P,0}^{\text{quanto}} = \bar{X} S_0 \exp[(r_f - q)T - \rho_{SX}\sigma_S\sigma_X T]\,, \tag{2.38}$$

where r_f is the foreign risk-free rate (Oesterreichische Nationalbank, 2004). The domestic option value is then obtained in Black forward form using $F_{P,0}^{\text{quanto}}$ and volatility σ_S,

discounting at r_d:

$$C_0^{\text{quanto}} = e^{-r_d T}\left[F_{P,0}^{\text{quanto}} N(d_1) - K N(d_2)\right], \tag{2.39}$$

$$P_0^{\text{quanto}} = e^{-r_d T}\left[K\, N(-d_2) - F_{P,0}^{\text{quanto}}\, N(-d_1)\right], \tag{2.40}$$

with d_1, d_2 defined as in (2.34) using $(F_{P,0}^{\text{quanto}}, \sigma_S)$ (Oesterreichische Nationalbank, 2004).

2.3.4 Option on a basket: moment-matching lognormal proxy

The structured strategies implemented in the tool are engineered so that the option payoff depends on the performance of the basket as a whole. Consistently with the buy-and-hold construction in Section 2.2.2, let N_i denote the number of shares of asset i purchased at inception. Define the domestic terminal price of asset i by:

$$P_T^{(i)} = \begin{cases} S_T^{(i)}\, X_T^{(i)}, & \text{Compo}, \\ S_T^{(i)}\, \bar{X}^{(i)}, & \text{Quanto}, \\ S_T^{(i)}, & \text{Domestic}. \end{cases} \tag{2.41}$$

The basket value at maturity is:

$$B_T = \sum_{i=1}^{n} N_i\, P_T^{(i)}. \tag{2.42}$$

A European basket call and put have payoffs:

$$\Pi_{\text{call}} = (B_T - K)^+, \qquad \Pi_{\text{put}} = (K - B_T)^+. \tag{2.43}$$

Because a sum of lognormal variables is not lognormal in general, the tool applies the standard moment-matching approximation described in Oesterreichische Nationalbank (2004). Under the domestic risk-neutral measure $\mathbb{Q}$, define the first two moments (Oesterreichische Nationalbank, 2004):

$$M_1 = \mathbb{E}^{\mathbb{Q}}[B_T], \qquad M_2 = \mathbb{E}^{\mathbb{Q}}[B_T^2]. \tag{2.44}$$

A lognormal proxy for B_T is then characterised by an effective basket forward and volatility (Oesterreichische Nationalbank, 2004):

$$F_B \equiv M_1, \qquad \sigma_B^2 = \frac{1}{T} \ln\left(\frac{M_2}{M_1^2}\right). \tag{2.45}$$

Let $F_{P,0}^{(i)}$ denote the domestic forward of $P_T^{(i)}$ under domestic discounting (computed

using the appropriate Compo/Quanto/Domestic convention). Let $\sigma_{P,i}$ denote the corresponding domestic log-volatility, and let ρ_{ij} denote the correlation between domestic log-returns of $P^{(i)}$ and $P^{(j)}$. Then:

$$M_1 = \sum_{i=1}^{n} N_i \, F_{P,0}^{(i)}, \tag{2.46}$$

$$M_2 = \sum_{i=1}^{n} \sum_{j=1}^{n} N_i N_j \, F_{P,0}^{(i)} F_{P,0}^{(j)} \exp(\rho_{ij} \sigma_{P,i} \sigma_{P,j} T) \, . \tag{2.47}$$

Using (F_B, σ_B), basket options are priced in Black forward form (Oesterreichische Nationalbank, 2004):

$$C_0 = e^{-r_d T} \Big(F_B N(d_1) - K N(d_2) \Big), \tag{2.48}$$

$$P_0 = e^{-r_d T} \Big(K N(-d_2) - F_B N(-d_1) \Big), \tag{2.49}$$

with

$$d_1 = \frac{\ln(F_B/K) + \frac{1}{2}\sigma_B^2 T}{\sigma_B \sqrt{T}}, \qquad d_2 = d_1 - \sigma_B \sqrt{T}. \tag{2.50}$$

Compo and Quanto inputs in (M_1, M_2)

The currency convention enters through the individual domestic forwards $F_{P,0}^{(i)}$ and domestic volatilities $\sigma_{P,i}$ used in (2.46)–(2.47). Under **Compo**, $P^{(i)} = S^{(i)} X^{(i)}$, so the domestic volatility includes equity, FX, and equity–FX covariance effects (as in (2.36)). Under **Quanto**, $P^{(i)} = \bar{X}^{(i)} S^{(i)}$ and the effect of equity–FX dependence is reflected in the quanto-adjusted forward (2.38) (Oesterreichische Nationalbank, 2004).

2.4 Payoff profiles used in the tool

All payoff profiles in the tool are defined as functions of the terminal basket level B_T, ensuring that the structured strategies and the buy-and-hold benchmark are expressed on the same underlying and are therefore directly comparable.

2.4.1 Capital-protected participation note (uncapped): ZCB + Basket Call

A capital-protected participation note combines a protected amount at maturity with uncapped participation in the basket upside:

$$V_T = F \; + \; \alpha \max(B_T - K, 0), \tag{2.51}$$

where F denotes the protected amount paid at maturity (implemented via a ZCB with present value given by (2.27)), K is the basket call strike, and α is the participation factor determined by the option budget. This follows the standard "bond + call" structure for capital-protected equity-linked notes (Oesterreichische Nationalbank, 2004).

2.4.2 Capital-protected participation note (capped): ZCB + Basket Call spread

A capped participation note replaces the uncapped call by a basket call spread:

$$V_T = F \; + \; \alpha\Big(\max(B_T - K_1, 0) - \max(B_T - K_2, 0)\Big), \qquad K_2 > K_1, \tag{2.52}$$

where K_1 is the lower strike and K_2 is the upper strike (cap). This is replicable using a long basket call at K_1 and a short basket call at K_2, together with a ZCB that delivers the protected amount F at maturity (Oesterreichische Nationalbank, 2004).

2.4.3 Protected risk-reversal note: ZCB + (Long Basket Call) + (Short Basket Put)

A protected risk-reversal note maintains a protected amount while financing a basket call by selling a basket put:

$$V_T = F \; + \; \alpha\Big(\max(B_T - K_c, 0) - \max(K_p - B_T, 0)\Big), \tag{2.53}$$

where K_c is the call strike and K_p is the put strike (typically $K_p < B_0 < K_c$). The combination of plain-vanilla basket options yields an asymmetric payoff that increases upside participation by accepting downside exposure below K_p, consistent with standard payoff engineering using calls and puts (Chia, 2010).

Chapter 3

Modelling Framework and Financial Engine

This chapter describes the the set of modelling assumptions and numerical procedures that transform user inputs (basket composition, maturity, product parameters, and market assumptions) into (i) simulated buy-and-hold outcomes and (ii) priced structured payoff strategies expressed on the same basket. The aim is to make the comparison problem operational: all strategies are evaluated over the same horizon and under the same underlying risk-factor dynamics, ensuring comparability by construction.

Beyond comparability, the engine is designed to support interpretability. Retail investors often perceive risk in terms of loss probability and payoff asymmetry rather than purely in terms of variance. Empirical evidence shows that loss probability and skewness can be stronger determinants of perceived risk than variance (Holzmeister et al., 2020). Moreover, disappointment-based preferences can rationalize stock market non-participation, even in the presence of an equity premium (Ang et al., 2005). Against this behavioural backdrop, the tool's key educational mechanism is to display terminal distributions and payoff shapes: it allows users to see how security design reshapes downside risk and upside participation, a channel consistent with evidence that capital-guarantee product design can foster risk-taking among previously risk-averse households (Calvet et al., 2023).

To bind the methodological description to the implementation, each major engine block is mapped to the corresponding code segment in Appendix A. The full script is reproduced in parts (Appendix A.1–A.14).

3.1 Market assumptions and inputs

To preserve computational efficiency and enable interactive exploration, the tool adopts a constant-parameter setting: risk-free rates, dividend yields, volatilities, and correlations

are treated as time-invariant over the horizon. This assumption is not meant to forecast markets; rather, it provides a controlled environment in which users can isolate the impact of key drivers (maturity, volatility, correlation, FX risk, and payoff terms) on risk–return outcomes.

The inputs are grouped into:

- **Investment horizon and capital:** (V_0, T) (investor UI in Appendix A.4, Listing A.4).
- **Basket definition:** asset set and weights $(w_i)_{i=1}^n$ (Section 2.2.1), implemented in Appendix A.4 (Listing A.4).
- **Equity parameters:** (μ_i, q_i, σ_i) used for performance simulation under $\mathbb{P}$ (Sections 2.1.2–2.1.6); defaults are defined in Appendix A.3 (Listing A.3) and can be overridden in Appendix A.5 (Listing A.5).
- **FX parameters (foreign assets):** $(r_d, r_{f,i}, \sigma_{X_i})$ and equity–FX correlations (Section 2.2.5); FX inputs appear in Appendix A.5 (Listing A.5).
- **Pricing parameters:** (r_d, q_i, σ_i) for Black–Scholes–Merton valuation under $\mathbb{Q}$ (Section 2.3.2); pricing primitives are implemented in Appendix A.10 (Listing A.10) and Appendix A.11 (Listing A.11).
- **Product parameters:** protected amount F, strikes, caps, and the currency convention (Compo/Quanto) (Sections 2.3.3 and 2.4); implemented in Appendix A.12 (Listing A.12) and wired in the main app in Appendix A.14 (Listing A.14).

3.2 Scenario engine for buy-and-hold outcomes

3.2.1 Correlated risk factors

The engine simulates a joint vector of Brownian shocks for all relevant risk factors (equities and foreign exchange) using the correlation matrix construction of Section 2.2.4. On each time step, correlated standard normals are generated via Cholesky factorization (Section 2.2.4), using (2.20)–(2.21). The correlation input panel is implemented in Appendix A.6 (Listing A.6). To ensure the matrix is Cholesky-admissible, a PSD repair step is applied as in Appendix A.7 (Listing A.7).

3.2.2 Equity and FX discretization under $\mathbb{P}$&$\mathbb{Q}$

Equity prices follow geometric Brownian motion (Section 2.1.2), and the tool uses the exact log-discretization summarized in Section 2.2.5. For each asset i, (2.25) updates

$\ln S_t^{(i)}$ using drift $(\mu_i - q_i)$ and volatility σ_i. For foreign assets, exchange rates are simulated according to (2.26), which corresponds to the benchmark FX process in Section 2.2.5. The joint simulator is implemented in Appendix A.8 (Listing A.8).

3.3 Buy-and-hold benchmark engine

3.3.1 Domestic valuation

Given weights (w_i) and initial capital V_0, the number of shares purchased at inception is computed as in (2.16). The domestic value of the buy-and-hold basket is then computed pathwise via (2.17), with terminal values given by (2.18). This construction ensures that both domestic and foreign assets are expressed in the investor's domestic currency (Section 2.2.2). The benchmark valuation is implemented in Appendix A.9 (Listing A.9) and is wired into the main app in Appendix A.13 (Listing A.13).

3.3.2 Return distribution

For each simulation path, the engine computes the simple and log returns of the benchmark as in (2.19). Importantly, even if each constituent is lognormal, the basket is a weighted sum of lognormals and is therefore not lognormal (Section 2.2.3). This motivates presenting the distribution empirically via Monte Carlo rather than imposing a normal approximation at the portfolio level. The histogram output is generated in the main app block (Appendix A.13, Listing A.13).

3.4 Pricing engine for structured strategies

Structured strategies are implemented in a transparent replication logic: a fixed-income leg (modelled as a zero-coupon bond) plus an option leg written on the basket level. This follows the Type 1 product focus of Vandenbroucke (2014), keeping payoffs directly linked to the underlying basket and therefore maximally comparable to buy-and-hold outcomes. The structured engine is implemented in Appendix A.12 (Listing A.12) and displayed in the main app in Appendix A.14 (Listing A.14).

3.4.1 Fixed-income leg

The protected component is modelled as a default-free zero-coupon bond with present value given by (2.27). The engine allocates part of the initial budget V_0 to this bond to deliver the protected amount F at maturity. This is implemented by `zcb_pv` in Appendix A.12 (Listing A.12).

3.4.2 Basket option valuation via moment matching

The option leg is written on the terminal basket level B_T defined in (2.42), with payoffs (2.43). Since B_T is a sum of lognormal variables, the tool applies the moment-matching lognormal proxy described in Section 2.3.4. Under $\mathbb{Q}$, the first two moments M_1, M_2 are computed using (2.46)–(2.47). These moments define an effective basket forward and volatility via (2.45). Basket option prices are then obtained in Black forward form using (2.50).

The Black forward primitive is implemented in Appendix A.10 (Listing A.10), and the moment-matching basket option implementation is in Appendix A.11 (Listing A.11).

3.4.3 Currency conventions in pricing: Compo vs Quanto

For foreign underlyings, the tool supports two standard conventions (Section 2.3.3). Under **Compo**, the domestic underlying is $P_t = S_t X_t$ (Equation (2.35)). Pricing uses the effective volatility (2.36) and domestic forward (2.37). Under **Quanto**, the payoff converts at a fixed rate and the equity–FX dependence enters through the quanto-adjusted forward (2.38), with valuation in (2.39)–(2.40). These adjustments are then propagated into the basket-moment computations through the inputs to (2.46)–(2.47) (Section 2.3.4).

In the implementation, the Compo/Quanto mapping enters through the construction of domestic forwards and factor loadings in Appendix A.11 (Listing A.11), and through the payoff-time basket construction in Appendix A.12 (Listing A.12).

3.5 From prices to implementable payoffs

Given the product definition (Section 2.4), the engine determines the scale of the option leg via an *option budget* constraint. Intuitively, after purchasing the ZCB, the remaining budget is used to buy (or sell) the required basket option combination. This determines the participation factor α appearing in (2.51), (2.52), and (2.53). The budget-to-participation logic (α) is implemented in Appendix A.12 (Listing A.12).

Finally, for each strategy, the engine maps the simulated terminal basket outcomes $B_T^{(m)}$ into terminal payoff values $V_T^{(m)}$ using the payoff equations in Section 2.4. These arrays of terminal values are the core objects passed to the visualization layer described in the next chapter, generated in Appendix A.14 (Listing A.14).

Chapter 4

Tool Design and Interface

This chapter documents how the prototype is used and how its interface operationalizes the core objectives of this thesis: (i) *comparability by construction* between a buy-and-hold benchmark and structured strategies written on the same basket, (ii) *transparent and traceable* inputs and transformations, and (iii) *visual, distribution-based* communication of risk and payoff asymmetry.

The emphasis on transparency and comparability is aligned with the requirements commonly highlighted for retail-investor digital tools: reliable inputs, traceability, and user-friendly access that supports cross-product comparison (European Commission, 2020). In addition, the interface is intentionally distribution-first: empirical evidence suggests that investors' perceived risk is strongly linked to loss probability, and histograms are a common way to communicate return distributions in decision settings (Holzmeister et al., 2020).

4.1 Use case

4.1.1 Reproducible reference scenario

To make the use case reproducible, all figures in this chapter are generated with a fixed random seed and fixed Monte Carlo settings. In the implementation, the random number generator is initialized with a constant seed (`7`) and the scenario engine uses a daily time step ($dt = 1/252$) and a fixed number of simulation paths (`n_paths = 500`). This makes the simulated paths and terminal distributions deterministic conditional on the same inputs, allowing a reader to replicate the same outputs when re-running the app under the same configuration.[1] The corresponding wiring is shown in Appendix A, Listing A.13.

[1] Exact numerical equality can still depend on software versions and numerical libraries; for the thesis use case, fixing seed and simulation settings is sufficient to reproduce the qualitative and near-identical quantitative results.

4.1.2 High-level layout: comparison-first design

The interface is organized around a comparison task: the left sidebar collects all inputs, while the main area displays (i) buy-and-hold scenarios and (ii) structured strategies evaluated on the same basket. The central design choice is to keep the benchmark visible while the user changes product parameters, reducing cognitive load and supporting "like-for-like" comparisons of payoff shapes and terminal return distributions. This emphasis on distributional and loss-related information aligns with evidence that perceived risk is often driven more by loss probability and asymmetry than by variance alone (Holzmeister et al., 2020), and with behavioural mechanisms such as disappointment-based preferences that can discourage equity participation (Ang et al., 2005).

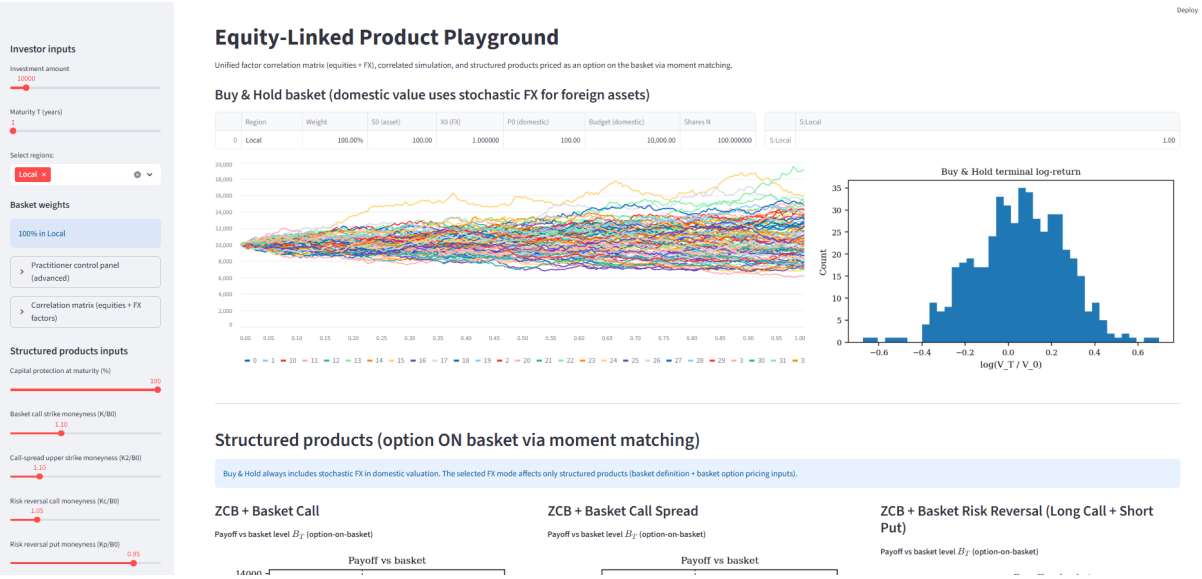

Figure 4.1: Dashboard overview (part 1). Sidebar inputs and the buy-and-hold benchmark: basket composition table, simulated path fan chart, and terminal log-return histogram.

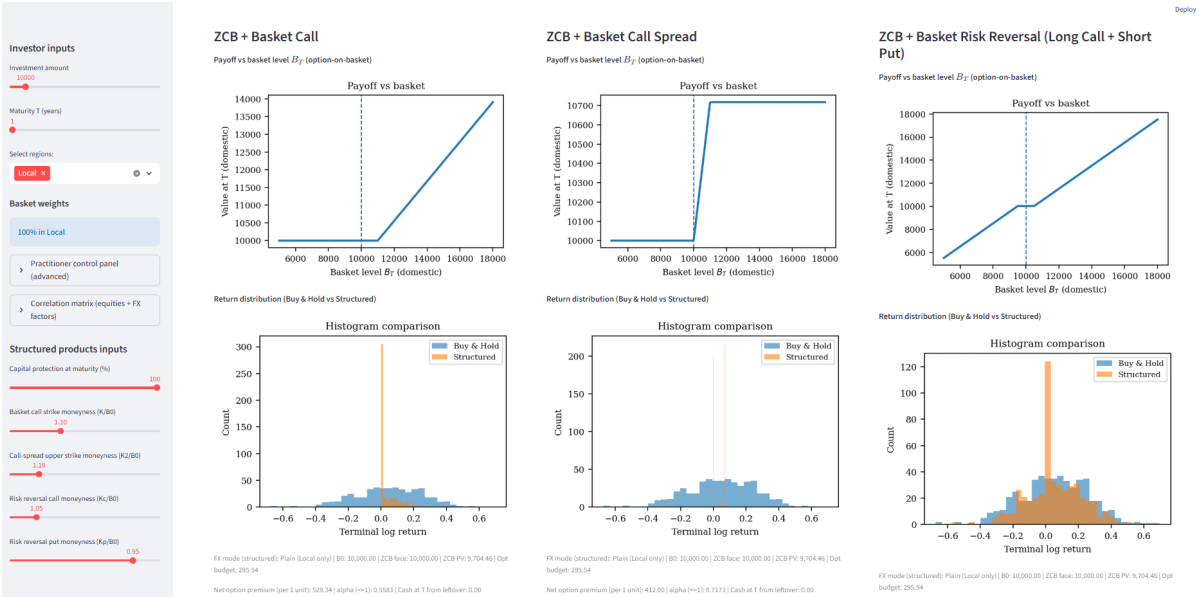

Figure 4.2: Dashboard overview (part 2). Structured strategies displayed side-by-side, each with a payoff curve as a function of the basket level and a histogram comparison against buy-and-hold.

4.1.3 Sidebar inputs: investor controls

The investor panel provides the minimum set of inputs required to define a clear experiment: investment amount, maturity, basket constituents, and weights. The panel is implemented in Appendix A, Listing A.4.

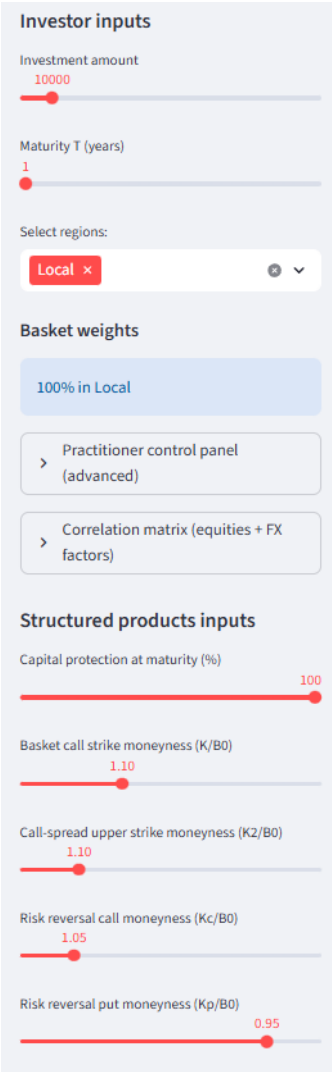

Figure 4.3: Investor side panel (core inputs). Investment amount, maturity, region selection, and basket weights.

Region selection and weights. Assets are presented as "regions" (Local, Foreign 1, Foreign 2) to keep the interface readable while still allowing the user to introduce cross-border features (foreign rates and FX risk). Once regions are selected, the tool enforces a valid long-only basket definition: weights are non-negative and sum to one. Internally, weights are converted into purchased share quantities N_i at inception (Appendix A, Listing A.9).

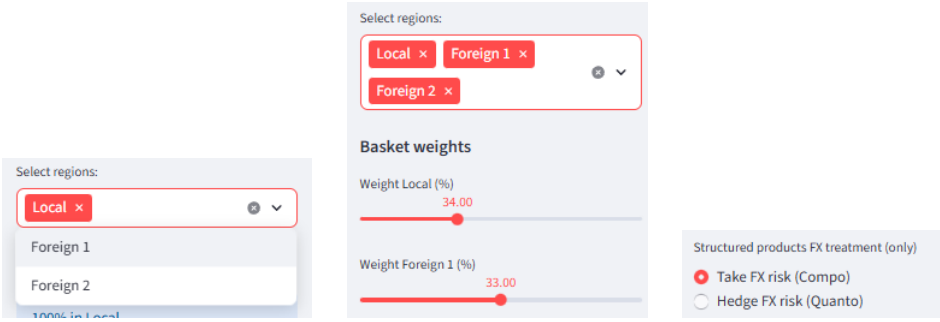

Figure 4.4: Input flow in the investor panel. Left: region selection menu. Middle: selected regions. Right: FX treatment toggle for structured strategies (Compo vs Quanto), shown once foreign regions are included.

Buy-and-hold versus structured FX convention. To highlight cross-border considerations, the tool separates the FX convention used for structured-product valuation from the buy-and-hold benchmark. The benchmark is always valued in domestic currency with stochastic FX for foreign holdings, so the benchmark distribution reflects unhedged cross-currency exposure. For structured strategies, the user can choose: (i) **Compo** (take FX risk in the payoff), or (ii) **Quanto** (fix conversion at inception). This toggle is propagated to the basket definition at maturity and to basket-option pricing inputs (Appendix A, Listings A.4 and A.11).

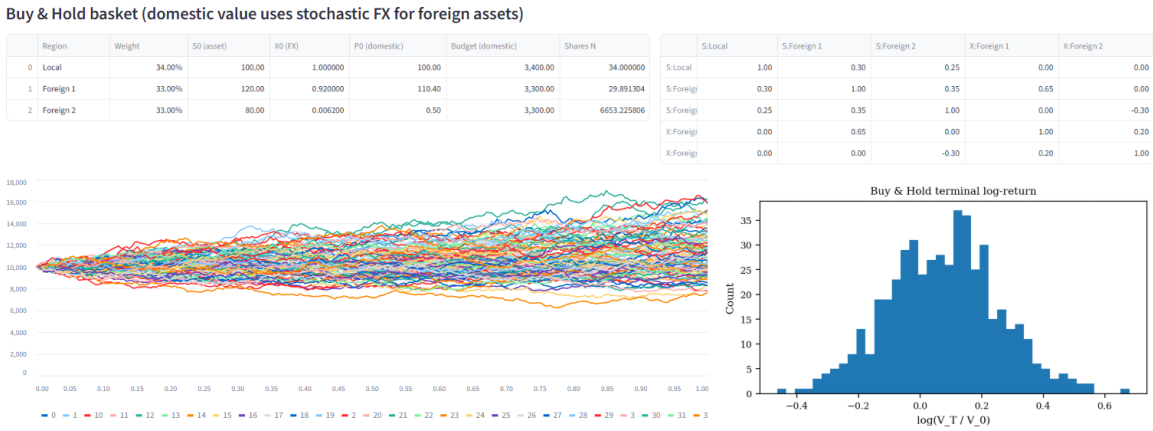

Figure 4.5: Basket summary in the main panel. The tool reports basket constituents, weights, domestic initial prices, budgets, and purchased share quantities N_i used in the buy-and-hold benchmark.

4.1.4 Advanced inputs: practitioner panel

To preserve a simple user experience while still supporting flexibility, advanced market inputs are hidden by default and exposed through an expandable practitioner panel (Appendix A, Listing A.5). The panel allows controlled experimentation with: (i) expected return parameters (used for scenario generation under $\mathbb{P}$), (ii) rates (used for discounting

and FX drift), and (iii) volatilities (equity and FX).

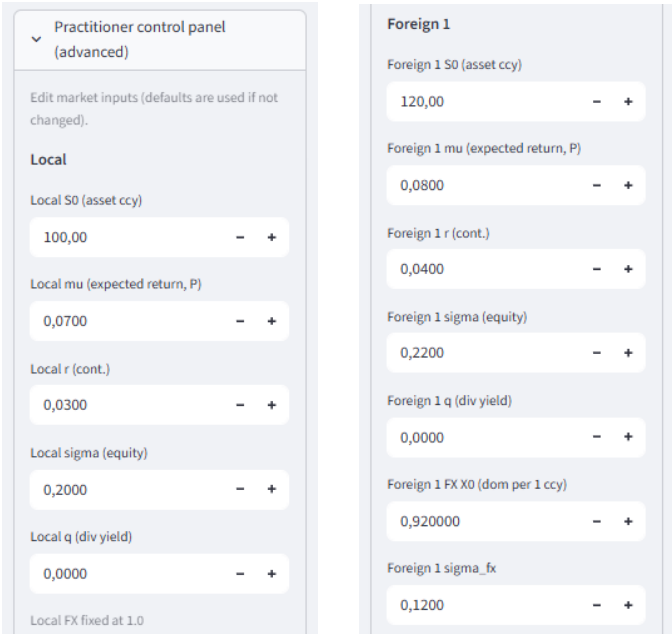

Figure 4.6: Practitioner panel. Advanced inputs for spot, expected return, rate, equity volatility, dividend yield, and FX parameters (for foreign regions).

4.2 Joint correlation matrix: equity and FX risk factors

A key strength of the prototype is that it exposes correlation assumptions transparently and applies them consistently to the entire risk-factor vector (equities and FX). The correlation sliders define the factor correlation matrix used for Cholesky-based generation of correlated shocks. If the matrix is not positive semi-definite, the tool applies a PSD repair step before factorization (Appendix A, Listings A.6 and A.7). This explicit exposure of modelling assumptions supports auditability and aligns with recommendations commonly made for investor-facing tools (European Commission, 2020)).

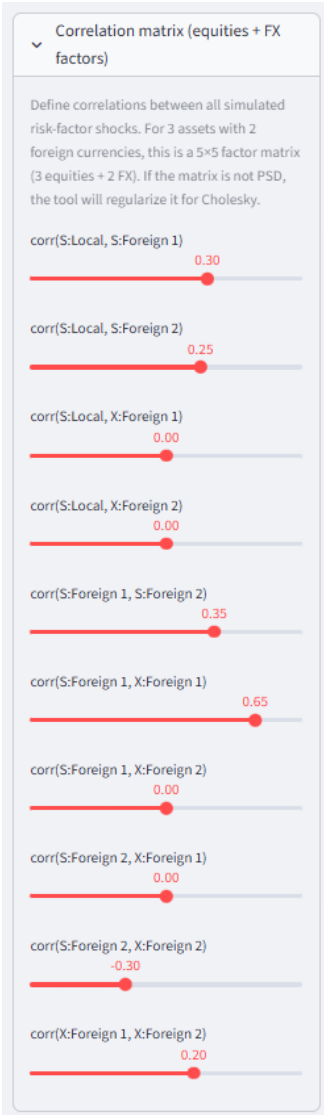

Figure 4.7: Unified correlation input. Correlations are specified across equity and FX shocks; the resulting factor matrix drives the correlated normal draws via Cholesky factorization (with PSD repair if required).

4.3 Structured product controls and outputs

Structured strategies are configured through a compact set of parameters: the protection level and strike moneyness inputs. These controls map directly into the replication logic (a zero-coupon bond plus an option combination) and therefore determine both the option budget available at inception and the implied participation factor α (Appendix A, Listing A.12).

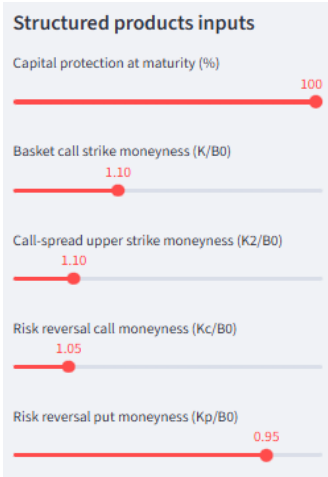

Figure 4.8: Structured-product inputs. Capital protection and strike moneyness parameters defining the option legs (basket call, call spread, and risk reversal).

4.3.1 What each structured-products input means

Let B_0 denote the initial basket value and B_T the terminal basket level in domestic currency. The structured-product controls affect outcomes through two linked mechanisms. First, the slider "Capital protection at maturity" sets the protected amount delivered by the bond at maturity. In the implementation this is defined as $F = V_0 \times \text{protection_pct}$. A higher protection level increases the present value invested in the zero-coupon bond, leaving a smaller residual budget to purchase the option leg. This budget split is tracked explicitly in the code through the variables `face`, `pv_zcb`, and `opt_budget` (Listing A.12).

Second, the moneyness inputs shape the option payoff by setting the relevant strikes as fractions of B_0. For the basket call, the strike is set by $K = (K/B_0) \cdot B_0$. For the call spread, the lower strike is anchored at $K_1 = B_0$ while the upper strike is set as $K_2 = (K_2/B_0) \cdot B_0$, which determines the cap level. For the protected risk reversal, the call and put strikes are set as $K_c = (K_c/B_0) \cdot B_0$ and $K_p = (K_p/B_0) \cdot B_0$. These strike choices determine where the payoff curve changes slope (the kinks), where it saturates in the capped structure, and where the strategy becomes exposed to downside due to the short put.

4.3.2 How the inputs change the graphs

Each strategy panel is designed to provide two complementary visual outputs. The first is a payoff curve $V_T(B_T)$, which shows the terminal payoff as a function of the basket level while holding all other parameters fixed. The second is a histogram comparison of terminal log returns, contrasting the structured strategy distribution with the buy-and-hold benchmark under the same simulated states.

Changes in the protection parameter primarily shift the balance between the bond

floor and upside participation. Increasing protection raises the payoff floor because F is larger, but it typically reduces the sensitivity of the payoff to B_T because less option budget implies a smaller participation factor α. Reducing protection has the opposite effect: more of V_0 is allocated to the option leg, the payoff curve becomes more responsive to the basket level, and the structured return distribution generally widens as the strategy becomes more equity-like.

Changes in strike moneyness primarily move payoff kinks and alter which market scenarios generate participation. Increasing the basket-call moneyness shifts the kink to the right, so upside participation begins only at higher basket realizations. Because a further out-of-the-money call is cheaper, α can increase for a fixed option budget, but the trade-off is that moderate upside scenarios no longer contribute to the option payoff. For the call spread, increasing K_2/B_0 raises the cap level and permits more upside, but it typically increases the net option cost of the spread. With a fixed budget, that higher cost can reduce α, so the payoff curve reflects a clear trade-off between higher upside capacity and weaker participation scaling. For the risk reversal, K_c/B_0 controls where upside begins, while K_p/B_0 controls how close the short put is to the money. Moving K_p upward (closer to-the-money) generally increases the premium received from selling the put, improving the financing of the call and potentially raising α, but it also expands the region of downside exposure below K_p. This tends to make left-tail risk more pronounced in the histogram and steepens the payoff slope in adverse scenarios.

4.3.3 How the inputs change the numbers below the plots

The numerical lines beneath each strategy summarize the replication bookkeeping that connects inputs to the displayed payoff. They report the initial basket level (`B0`), the bond face value and its present value (`ZCB face` and `ZCB PV`), and the resulting residual funds available for derivatives (`opt_budget`). They also display the basket-option valuation output (`net_premium`) and the implied participation factor, computed in budget terms as $\alpha = \min\{1, \texttt{opt_budget}/\texttt{net_premium}\}$ (with adjustments in cases where the option combination is a net credit, which can occur for some risk-reversal settings). Any budget not spent on options is carried as cash and grown at the domestic risk-free rate to maturity (`cash_T`). These quantities are computed in Listing A.12, using basket option premia produced by the moment-matching pricing routine (Listing A.11).

4.3.4 Illustrative parameter changes (screenshots)

Figures 4.9–4.12 illustrate how the interface supports controlled experimentation: changing a single input while holding others fixed produces immediate and interpretable changes in both payoff shape and the terminal return distribution.

Figure 4.9: Effect of reducing capital protection. Lower protection decreases the bond face value F, increases the option budget, and typically raises α, making the payoff curve more responsive to B_T and widening the structured return distribution.

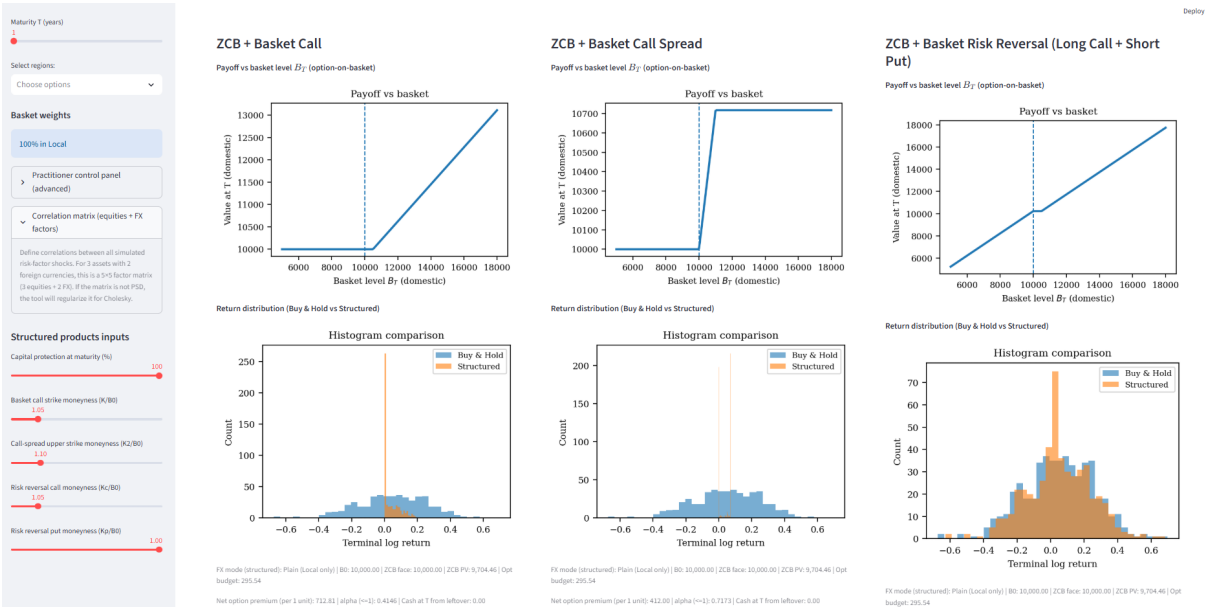

Figure 4.10: Effect of changing basket call moneyness (K/B_0). Increasing moneyness shifts the payoff kink to higher basket levels, concentrating outcomes near the protected region for moderate B_T while preserving upside only for larger rallies.

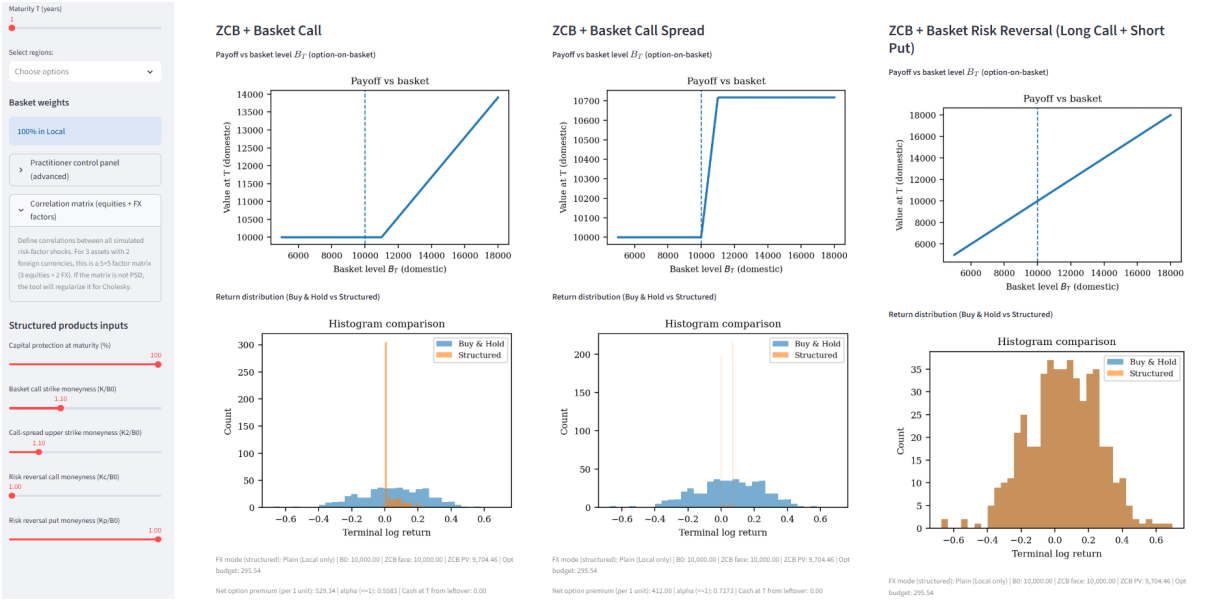

Figure 4.11: Effect of changing risk-reversal call moneyness (K_c/B_0). A higher K_c delays upside participation (kink shifts right) and changes the net premium and α through the call's cost.

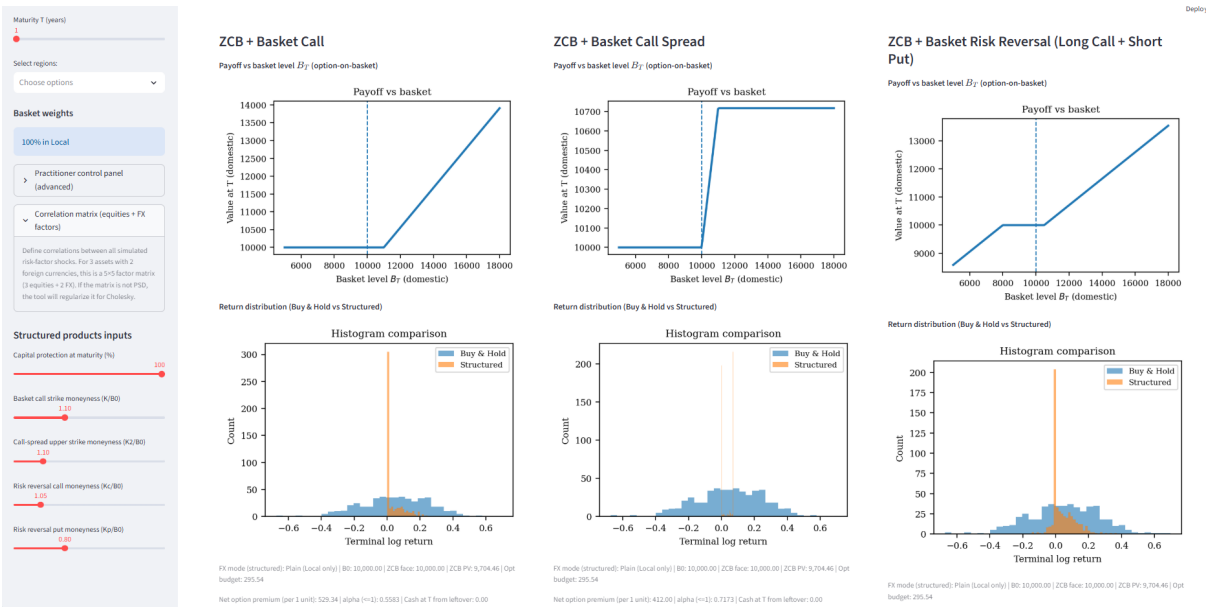

Figure 4.12: Effect of changing risk-reversal put moneyness (K_p/B_0). Moving K_p alters the downside exposure region and the premium received from the short put, reshaping left-tail risk in the structured distribution and the payoff slope below K_p.

4.4 Implementation traceability: linking text to code

To support transparency and auditability, each interface block corresponds to a clearly separated implementation component in the appendix: investor inputs (Listing A.4), practitioner inputs (Listing A.5), correlations (Listings A.6–A.7), scenario generation

(Listing A.8), buy-and-hold valuation (Listing A.9), and basket-option / structured strategy valuation (Listings A.11 and A.12). This structure makes the transformation from modelling assumptions to displayed outputs explicit and inspectable, which is particularly important for retail-facing tools (European Commission, 2020).

Chapter 5

Discussion

This thesis developed a focused prototype that operationalizes key principles proposed in the EU discussion on improving retail investment decision-making through digital tools and centralized information access (European Commission, 2020). Within a deliberately limited scope—Type 1 equity-linked structured products and a parsimonious stochastic modelling framework—the contribution is to make the retail *comparison problem* tangible: buy-and-hold and multiple structured payoffs are evaluated over the same horizon, on the same basket, and under the same joint risk-factor scenarios.

5.1 Overview and Objectives

The prototype contributes along three dimensions that recur in the policy and behavioural motivations of this thesis.

In terms of market transparency, the prototype effectively addresses the systemic lack of product comparability by embedding a 'benchmarking-by-design' framework. A central barrier identified in the policy landscape is that retail investors struggle to compare products across channels and disclosures (Council of the European Union, 2023; European Commission, 2020). The tool addresses this by design: it anchors the analysis on a buy-and-hold benchmark in the same basket and compares it directly to structured strategies written on that same basket. This avoids the typical ambiguity where products are compared across different underlyings, horizons, or risk assumptions, and it forces all outcomes into a common metric (terminal value and return distribution in domestic currency).

Beyond mere information provision, the tool facilitates deeper interpretability by deconstructing complex payoff mechanics into their constituent fundamental components. A second friction is that disclosure does not automatically translate into usable understanding. The tool therefore makes product engineering explicit: each strategy is decomposed into a protected amount (ZCB) plus an option budget and a basket-option leg. Impor-

tantly, the interface communicates risk not only through point estimates but through payoff curves and terminal distributions. This approach matches behavioural evidence that perceived risk is often driven by loss probability and asymmetry rather than variance alone (Holzmeister et al., 2020), and connects naturally to disappointment-based preference mechanisms that can rationalize non-participation in equities even when premia exist (Ang et al., 2005).

Finally, the prototype serves as a rigorous analytical environment for exploring the often-overlooked impact of cross-border currency risk on retail investment outcomes. By modelling foreign assets jointly with FX and enabling Compo versus Quanto conventions, the prototype highlights how cross-currency exposure changes both benchmark outcomes and structured product valuation. This responds to a practical feature of EU retail portfolios: diversification frequently implies cross-border exposure, but the FX component is often poorly understood in retail contexts. The tool makes this explicit via a single switch that immediately changes the payoff interpretation and the distribution outcomes.

5.2 How the prototype responds to the market issues in perspective

The EU retail-investment agenda emphasizes participation, trust, and better use of household savings (Council of the European Union, 2023; European Commission, 2020). This thesis does not claim to solve these structural issues. Instead, it demonstrates a narrow but relevant mechanism: *transparent comparison reduces informational friction*. Specifically, the tool shows how structured products can be understood as distribution-shaping instruments rather than opaque "complex products". This is consistent with evidence that capital-protection designs can encourage risk taking among initially more risk-averse households (Calvet et al., 2023). In other words, the prototype illustrates how product design can address behavioural concerns (downside probability, payoff asymmetry) while maintaining access to market upside, and it provides a format that could be compatible with a future "products hub" logic (European Commission, 2020).

The contribution should therefore be interpreted as a *proof of concept*: it shows that a relatively simple interface, grounded in transparent modelling blocks, can still be powerful when it consistently connects assumptions → pricing → payoffs → distributions.

5.3 Limitations and modelling assumptions

The conclusions above must be interpreted in light of the prototype's scope and assumptions.

While the tool provides a robust framework for core strategies, its current utility is

bounded by a focus on Type 1 structures, as defined by pragmatic complexity classifications. The implementation focuses on Type 1 structures (bond ± plain-vanilla option legs) following the pragmatic complexity classification (Vandenbroucke, 2014). This choice supports clarity and tractability, but excludes important retail market segments such as autocallables, callable notes, and many path-dependent designs (Types 2–4). As a result, the tool's current outputs should not be interpreted as representative of the full structured product universe.

From a modelling perspective, the prototype relies on idealized market dynamics that prioritize baseline transparency over the capturing of complex empirical phenomena. The engine assumes constant rates, volatilities, dividends, and correlations over the horizon. Equity and FX are modelled with lognormal dynamics (GBM) and joint Gaussian shocks. These assumptions are standard as a baseline (Hull, 2018), but they are not intended as forecasting models and they abstract away from regime shifts, volatility clustering, jumps, and time-varying correlation—all of which can be relevant in stressed markets.

The use of moment-matching approximations for basket option valuation introduces a specific layer of model risk that must be considered when interpreting quantitative outputs. Basket options are valued via a moment-matching lognormal proxy. This supports speed and transparency, but introduces model risk: the approximation is generally accurate for some parameter ranges and less accurate for others (e.g., extreme skewed baskets, strong heterogeneity, or long maturities). Therefore, the displayed prices and α values should be treated as internally consistent within the framework rather than as tradable quotes.

Finally, the engine operates as a theoretical pricing and comparison tool, intentionally omitting several real-world frictions that significantly influence net investor welfare. The prototype models the bond as default-free and does not incorporate issuer credit risk, liquidity premia, funding spreads, fees, bid–ask spreads, taxes, or early exit constraints. In real retail products, these elements can materially affect realized performance and investor welfare. The tool is best interpreted as a payoff-engineering and comparability engine rather than a complete product suitability or profitability evaluator.

5.4 Future perspectives

The thesis suggests several natural extensions that would broaden relevance while preserving the comparison-first design. To transition the prototype from a theoretical framework to an operational tool, a primary development path involves the systematic integration of real-time market data and regulatory disclosure inputs. A direct next step is to link the engine to observable market data (rates, implied vol surfaces, FX vols, correlations) and to structured product disclosures (e.g., termsheets / KIDs), so that scenarios and prices can be calibrated rather than set manually. This would move the prototype closer to

the operational vision of a centralized products hub (European Commission, 2020) and would enable systematic cross-product comparisons under harmonized inputs.

A second step would be to broaden the engine's architectural scope to accommodate more complex Type 2–4 structures, such as autocallables, without sacrificing user interpretability A second step is to extend to selected Type 2–4 structures (e.g., autocallables) (Vandenbroucke, 2014). This would require additional modelling blocks (path dependence, early termination logic) and careful interface design to avoid overwhelming the user. A pragmatic approach would be to add products incrementally and maintain the same narrative: explicit mechanisms, payoff curve, and distribution comparison.

Enhancing the realism of the engine's outputs could be achieved by incorporating institutional frictions and more granular, investor-centric risk metrics. To strengthen realism, the tool could incorporate fees, bid–ask spreads, and issuer credit risk (e.g., credit spread discounting on the bond leg), and display additional metrics aligned with retail concerns: probability of capital loss, expected shortfall, downside semi-variance, and scenario-based drawdown summaries. These outputs would complement the existing histogram view while staying consistent with behavioural evidence on loss-focused risk perception (Holzmeister et al., 2020).

Finally, the ultimate utility of the prototype should be validated through empirical research to measure its actual impact on retail investor decision-making. Finally, the prototype's central promise is improved interpretability and decision support. A structured follow-up would test whether the interface changes risk understanding, confidence, or product selection in controlled experiments, and whether distribution-based visualization reduces misunderstandings relative to standard static disclosure. This would directly connect the engineering contribution of the thesis to measurable investor outcomes.

Chapter 6

Conclusion

Within its limited but intentional scope, this thesis demonstrates that the integration of transparent modelling blocks with a comparison-first user interface can effectively operationalize the core policy pillars of neutrality, usability, and comparability (European Commission, 2020). By shifting the analytical focus from isolated product features to relative performance against a buy-and-hold benchmark, the prototype addresses the information asymmetry that often characterizes the retail investment landscape. The developed engine should be understood as a modular foundation rather than a static end-point. It proves that there is a feasible, technically sound pathway to translate complex payoff engineering into an interactive, auditable, and cognitively accessible decision aid. While the current implementation focuses on Type 1 structures, the framework establishes the necessary logic to eventually scale the product universe and incorporate greater market realism. Ultimately, this work serves as a proof-of-concept for a more democratic investment environment, where sophisticated financial engineering is no longer a barrier to investor understanding, but a transparent tool for wealth management.

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